Synchronous concurrent refinement algebra and justness

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Overview

- Brief overview of synchronous concurrent refinement algebra
- Justness – a form of fairness

Structure of concurrent program algebra

- Concurrent refinement algebra ($\sqcap$, $\sqcup$, abort, magic, ; , nil, $\|$, skip, $\Cap$, chaos)
- Plus tests – a subset of commands that forms a boolean algebra (like Kozen's KAT)
- Plus atomic steps – a subset of commands that forms a boolean algebra
- Program/environment steps – partitions atomic steps
- Relational instantiation

Aczel traces

Every step of parallel synchronises steps of the two processes

$$\begin{array}{ll} \mathcal{E}(\sigma_0,\sigma_1),\Pi(\sigma_1,\sigma_2),\mathcal{E}(\sigma_2,\sigma_3),\mathcal{E}(\sigma_3,\sigma_4),\mathcal{E}(\sigma_4,\sigma_5) & \| \\ \mathcal{E}(\sigma_0,\sigma_1),\mathcal{E}(\sigma_1,\sigma_2),\Pi(\sigma_2,\sigma_3),\mathcal{E}(\sigma_3,\sigma_4),\Pi(\sigma_4,\sigma_5) & = \\ \mathcal{E}(\sigma_0,\sigma_1),\Pi(\sigma_1,\sigma_2),\Pi(\sigma_2,\sigma_3),\mathcal{E}(\sigma_3,\sigma_4),\Pi(\sigma_4,\sigma_5) & \end{array}$$

Some primitive commands

For a binary relation $r \subseteq \Sigma \times \Sigma$ on states

$\pi(r)$ can perform any single program step $\Pi(\sigma, \sigma')$ for $(\sigma, \sigma') \in r$

$\epsilon(r)$ can perform any single environment step $\mathcal{E}(\sigma, \sigma')$ for $(\sigma, \sigma') \in r$

For example,

- $\boldsymbol{\pi}(\mathrm{id})$ is a single stuttering program step
- $\boldsymbol{\pi} \mathrel{\widehat{=}} \boldsymbol{\pi}(\mathrm{univ})$ can perform any single program step
- $\boldsymbol{\epsilon} \mathrel{\widehat{=}} \boldsymbol{\epsilon}(\mathrm{univ})$ can perform any single environment step
- $\boldsymbol{\pi}(\varnothing) = \boldsymbol{\epsilon}(\varnothing) =$ magic is infeasible

Operators

$c \sqcap d$ non-deterministic choice (lattice meet)

$c \sqcup d$ lattice join

$c \,;\, d$ sequential composition (elided to $c\,d$ below)

$c \parallel d$ parallel composition

$c \Cap d$ weak conjunction

Combining primitive commands with parallel and weak conjunction

$$
\begin{aligned}
\pi(r_1) \parallel \pi(r_2) &= \text{magic} \\
\pi(r_1) \parallel \epsilon(r_2) &= \pi(r_1 \cap r_2) \\
\epsilon(r_1) \parallel \epsilon(r_2) &= \epsilon(r_1 \cap r_2) \\
\pi(r) \parallel \text{abort} &= \text{abort} \\
\epsilon(r) \parallel \text{abort} &= \text{abort}
\end{aligned}
$$

$$
\begin{aligned}
\pi(r_1) \Cap \pi(r_2) &= \pi(r_1 \cap r_2) \\
\pi(r_1) \Cap \epsilon(r_2) &= \text{magic} \\
\epsilon(r_1) \Cap \epsilon(r_2) &= \epsilon(r_1 \cap r_2) \\
\pi(r) \Cap \text{abort} &= \text{abort} \\
\epsilon(r) \Cap \text{abort} &= \text{abort}
\end{aligned}
$$

Iteration

Iteration zero or more times, c^{ω}, allows finite iteration, $c^{\star}$, or infinite iteration, c^{∞}

$$c^{\omega} = c^{\star} \sqcap c^{\infty} \tag{1}$$

Examples

$\pi^{\star}$ can perform a finite number of program steps

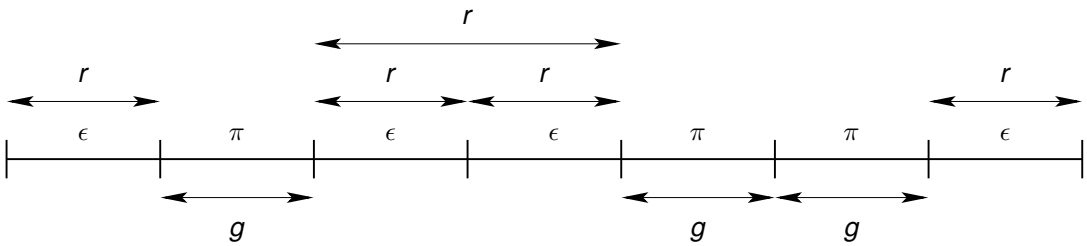
$(\pi \sqcap \epsilon)^{\star}$ can perform a finite number of steps

ϵ^{∞} performs an infinite sequence of environment steps

skip is the identity of parallel and chaos is the identity of weak conjunction

$$\begin{aligned} \text{skip} &\mathrel{\widehat{=}} \epsilon^{\omega} \\ \text{chaos} &\mathrel{\widehat{=}} (\pi \sqcap \epsilon)^{\omega} \end{aligned}$$

Rely/guarantee



Guarantee and rely

For relations g and r

$$\begin{aligned} \text{guar } g &\mathrel{\widehat{=}} (\pi(g) \sqcap \epsilon)^\omega \\ \text{demand } r &\mathrel{\widehat{=}} (\pi \sqcap \epsilon(r))^\omega \\ \text{rely } r &\mathrel{\widehat{=}} (\text{demand } r)\,(\text{nil} \sqcap \epsilon(\bar{r})\,\text{abort}) \end{aligned}$$

For example, $c \Cap (\text{guar } g) \Cap (\text{rely } r)$ imposes a guarantee of g on c and assumes the environment steps satisfy r.

Rely-guarantee specification of a component

- s is a set of natural numbers
- $mults(i)$ is the set of all multiples of i (excluding i)
- Specification

$$(\text{guar}\ \ s' \subseteq s \land s - s' \subseteq mults(i)) \Cap (\text{rely}\ \ s' \subseteq s) \Cap \{s = 2\,..\,N\}\,[s' \cap mults(i) = \varnothing]$$
$$\sqsubseteq$$
$$rem\_mult(i)$$

Termination

The command term allows only a finite number of program steps but does not rule out infinite pre-emption by its environment.

$$\text{term} \mathrel{\widehat{=}} \epsilon^{\omega}\,(\pi\,\epsilon^{\omega})^{\star} \tag{2}$$

The refinement

$$\text{term} \sqsubseteq c$$

states that c terminates if the environment does not interrupt it forever, e.g.

$$\text{term} \sqsubseteq x := 1$$

Imposing justness (a form of fairness)

The process just allows any behaviour, except pre-emption by its environment forever,

$$\text{just} \mathrel{\widehat{=}} \boldsymbol{\epsilon}^\star \, (\boldsymbol{\pi} \, \boldsymbol{\epsilon}^\star)^\omega \tag{3}$$

It requires all contiguous subsequences of environment steps to be finite.

Just termination

$$\begin{aligned} \mathsf{term} \mathbin{⫛} \mathsf{just} &= \epsilon^{\omega} \, (\pi \, \epsilon^{\omega})^{\star} \mathbin{⫛} \epsilon^{\star} \, (\pi \, \epsilon^{\star})^{\omega} \\ &= \epsilon^{\star} \, (\pi \, \epsilon^{\star})^{\star} \\ &= (\pi \sqcap \epsilon)^{\star} \end{aligned}$$

Hence if

$$\mathsf{term} \sqsubseteq c$$

then

$$\mathsf{term} \mathbin{⫛} \mathsf{just} \sqsubseteq c \mathbin{⫛} \mathsf{just}$$

i.e. just-execution of c gives strong termination.

Just execution

A process representing just-execution of a process c is represented by

$$c \Cap \mathrm{just}$$

Because just never aborts, any aborting behaviour of $c \Cap \mathrm{just}$ arises solely from c.

... just execution

Execution of the process,

$$\textbf{do}\ \text{true} \rightarrow y := y + 1\ \textbf{od} \tag{4}$$

with y initially zero may never set y to 7.
In contrast,

$$\textbf{do}\ \text{true} \rightarrow y := y + 1\ \textbf{od}\ \Cap\ \textbf{just} \tag{5}$$

rules out infinite pre-emption by its environment and hence ensures that eventually y becomes 7 (or any other natural number).

Unjust parallel

For a parallel composition, $c \parallel d$, the execution of c may be pre-empted forever by the execution of d, or vice versa.

$$x := 1 \parallel \text{do}\, x \neq 1 \rightarrow y := y + 1\, \text{od} \tag{6}$$

Just parallel

$$((x := 1) \Cap \text{just}) \parallel_e (\text{do } x \neq 1 \rightarrow y := y + 1 \text{ od} \Cap \text{just}) \tag{7}$$

The interpretation of $\parallel_e$ is that termination of one process of a parallel composition reduces the composition to the other process, i.e. $\text{nil} \parallel_e c = c$. One may define just-parallel as follows.

$$c \parallel_j d \;\widehat{=}\; (c \Cap \text{just}) \parallel_e (d \Cap \text{just}) \tag{8}$$

Note that

$$(c \parallel_e d) \Cap \text{just} \;\sqsubseteq\; c \parallel_j d \tag{9}$$

but the reverse refinement doesn't hold in general because $(c \parallel_e d) \Cap \text{just}$ doesn't rule out c being pre-empted forever by d (or vice versa) within the parallel.

Parallel with synchronised termination

For parallel with synchronised termination one only has

$$\mathsf{nil} \parallel \mathsf{nil} = \mathsf{nil}$$

but not the more general property of parallel with early termination.

$$\mathsf{nil} \parallel_e c = c$$

For parallel with synchronised termination,

$$(x := 1 \Cap \mathsf{just}) \parallel (\mathsf{do}\,\mathsf{true} \to y := y + 1\,\mathsf{od} \Cap \mathsf{just}) \tag{10}$$

is infeasible.

$$c \parallel_e d \;\widehat{=}\; (c\,\mathsf{skip}) \parallel (d\,\mathsf{skip}) \tag{11}$$

The definition of just-parallel then expands as follows.

$$c \parallel_j d \widehat{=} ((c \Cap \mathsf{just})\,\mathsf{skip}) \parallel ((d \Cap \mathsf{just})\,\mathsf{skip}) \tag{12}$$

That imposes justness on c until it terminates, and similarly for d.

Applying to process algebras

The above approach can also be applied to CSP-style processes.

- $\Pi(a)$ is interpreted as an atomic event or action a
- $\mathcal{E}(a)$ is a corresponding environment event
- $\pi(A)$ allows any event $\Pi(a)$ for any $a \in A$
- $\epsilon(A)$ allows any environment event $\mathcal{E}(a)$ for any $a \in A$.
- $\pi(A) \parallel \pi(B) = \pi(A \cap B)$

Summary

- Aczel's parallel operator is synchronous, similar to Milner's SCCS
- Using synchronous atomic steps algebra one can build a rely/guarantee theory
- Justness can be imposed on termination and parallel composition

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